Where to Act or Which Way to Move? Diagnosing Contact and Direction in Deformable Object Manipulation

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Abstract

Hybrid manipulation policies jointly choose a contact location p and a motion command u, so successful one-step behavior does not reveal whether progress comes from choosing *where* to act, which planar direction to move, or their interaction. We introduce a preregistered complete-factorial diagnostic for quasi-static deformable linear object (DLO) manipulation: for each canonicalized rope state we execute all 8×8 combinations of arc-length contact positions and fixed-magnitude planar motion directions and compare the value of selecting contact alone, direction alone, and both. The claim is parameterization-specific: we compare contact-location choice with fixed-magnitude planar-direction choice, not "where" versus "how" in general. On a development-split pilot of 40 valid C0 state tables (2,560 analyzed action cells) across three tasks, the descriptive contrast $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$ is +0.041 [+0.029, +0.053] for straightening, -0.034 [-0.073, +0.005] for single-bend shaping, and -0.130 [-0.161, -0.100] for endpoint repositioning (simultaneous 95% intervals). Applying the preregistered labeling rule descriptively yields contact-dominant, balanced/inconclusive, and direction-dominant labels, respectively. However, 42 of 192 state–candidate tables were invalid, mainly due to a canonicalization re-placement artifact: no candidate execution regime passed the preregistered invalid-table-rate guard, C0 (the first-priority regime) was *not* formally selected and is reported descriptively, and no hidden-split confirmation or clean reproduction was run. The contribution is a transparent, guard-audited action-attribution protocol and pilot evidence, not confirmatory conclusions.

1. Introduction

Deformable linear objects (DLOs)—ropes, cables, wires—remain hard for robots: state is high-dimensional and dynamics are contact-mediated and underactuated (Sanchez et al., 2018; Zhu et al., 2022). Practical systems therefore act through *hybrid* actions: a spatial choice of contact location paired with a motion command, as in dense per-point critic and affordance maps with a motion head (Zhou et al., 2023; Mo et al., 2021; Wu et al., 2023; Zeng et al., 2021). These architectures optimize the pair jointly and benchmarks score only the joint outcome, so success does not reveal whether it came from selecting *where* to act (contact location), which planar direction to move, or their pairing—yet that attribution decides whether the next unit of modeling effort should buy denser contact maps, richer motion generation, or joint reasoning.

We measure the attribution by direct intervention rather than by probing a trained policy, in a deliberately small setting: one-step, quasi-static DLO manipulation. For each canonicalized rope state we execute *every* cell of an 8×8 grid of contact positions $\times$ planar motion directions in a GPU-parallel rod simulator (Cao et al., 2026) and record normalized one-step progress; the value of knowing *where*, the direction, or *both* is then computable per state with no model in between—classical factorial experimentation (Montgomery, 2017) applied to an action space.

Narrow claim boundary. The two factors are *not* symmetric in expressiveness: the contact factor covers the rope's full normalized arc length at 8 points, while the motion factor covers only 8 planar directions at a fixed displacement magnitude—no magnitude, vertical, or rotational freedom. Equal 8×8 cardinality equalizes search budget, not representational completeness. All results therefore compare *contact-location choice versus fixed-magnitude planar-direction choice*, not "where versus how" in general; observed dominance of either factor can partially reflect this asymmetry.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Contributions. (i) A preregistered (Nosek et al., 2018) factorial design for one-step DLO action attribution with explicit validity guards, a complete-table-or-invalid policy separating invalid runs from valid null results, and effect-blind fixed-priority candidate selection. (ii) A pilot execution on three tasks with complete raw tables, controls, and cluster-bootstrap uncertainty, suggesting factor importance reverses across tasks. (iii) An honest account of a qualification failure: a canonicalization re-placement artifact invalidated 42 of 192 state–candidate tables and tripped the invalid-table-rate guard, so no candidate regime was formally selected; all numbers are descriptive development-split pilot evidence (deviations in Section 3.4).

2. Related work

DLO and deformable manipulation. DLOs are a canonical testbed for manipulation beyond rigid-body assumptions (Sanchez et al., 2018; Zhu et al., 2022). Prior work servos object shape through estimated deformation models (Berenson, 2013; Navarro-Alarcon et al., 2016; Yang et al., 2023), learns rope dynamics or policies from interaction (Nair et al., 2017; Wu et al., 2020; Chi et al., 2022), and standardizes evaluation in simulation benchmarks (Lin et al., 2021; Cao et al., 2026). We propose no new method; we measure how much one-step progress each half of a hybrid action carries—a quantity these evaluations do not report.

Dense contact affordances and spatial action selection. A prominent architectural family scores *where* to act densely over the object or scene: Where2Act (Mo et al., 2021), DualAfford (Zhao et al., 2023), dense affordances for deformables (Wu et al., 2023), HACMan's per-point critic maps (Zhou et al., 2023), and Transporter-style spatial action maps for cables and fabrics (Zeng et al., 2021; Seita et al., 2021). These designs commit capacity to the *where* factor by construction; we ask when that commitment is empirically justified for DLOs.

Hybrid and parameterized action selection. Discrete–continuous action spaces are formalized as parameterized-action MDPs (Masson et al., 2016) with dedicated architectures (Hausknecht & Stone, 2016); robotic instantiations couple a discrete pick with continuous motion parameters, e.g. TossingBot (Zeng et al., 2020). This literature optimizes the joint choice but does not quantify each factor's marginal value; our factorial tables provide exactly that decomposition for one primitive.

Attribution and evaluation rigor. Our estimands are classical factorial quantities (Montgomery, 2017) estimated with cluster bootstrap (Efron & Tibshirani, 1993); the reporting responds to reliability failures in empirical RL (Henderson et al., 2018; Agarwal et al., 2021) and preregistration practice (Nosek et al., 2018).

3. Study design

3.1. Environment, states, and primitive

We use a GPU-parallel discrete-rod simulator built on a differentiable rod solver (Cao et al., 2026) with a 32-vertex rope. States are *canonicalized quasi-static* configurations: an analytic initial centerline is placed, velocities are zeroed, and the rope is settled to a velocity threshold of 10^{-3} (capped at 10^4 steps). The primitive is grasp–move–release–settle at a chosen vertex; grasp realism is disabled so attachment is deterministic. Placement error after canonicalization is guarded at $\leq 10^{-4}$ m per table (Section 3.4).

Tasks are t1a *straighten*, t1b *single bend*, and t1c *endpoint reposition*, with stratified initial shapes ({u_bend, s_curve, random_smooth}; additionally {straight} for t1b/t1c). Task error D is a correspondence-L_2 distance to a goal template; a state is eligible only if $D_{\text{before}} \geq 0.10$ (twice the success threshold). Action quality is normalized one-step progress $Y(p,u) = \big(D_{\text{before}} - D_{\text{after}}(p,u)\big)/\max(D_{\text{before}}, 10^{-8})$, where p is the contact location and u the motion direction: $Y = 1$ solves the task in one step, $Y = 0$ is no progress, and $Y < 0$ makes the state worse.

3.2. Factorial grid, candidates, and experimental units

Contacts are 8 normalized arc-length positions $\sigma_i = i/7$ mapped to the nearest native vertex; motions are 8 planar directions at $45°$ increments. A *candidate execution regime* fixes the primitive's remaining free parameters—displacement magnitude and lift height—for an entire run: C0 $(0.10\,\text{m}, \text{low})$, C1 $(0.05\,\text{m}, \text{low})$, C2 $(0.10\,\text{m}, \text{high})$, C3 $(0.05\,\text{m}, \text{high})$. Table 1 gives the unit hierarchy and where invalid tables are excluded. Candidate selection is *fixed-priority after hard validity* (C0→C1→C2→C3): the first candidate passing the hard validity guards of Section 3.4 is selected; effect size, sign, and p-values are never consulted. In this pilot *no candidate passed* (Section 4); C0, the preregistered first-priority candidate, is reported descriptively.

3.3. Estimands: reading an 8×8 table

Each complete table answers four increasingly informed queries: a uniformly random cell earns the table mean in expectation; choosing only the *contact* well earns at best the best *row* mean; choosing only the *direction* well at best the best *column* mean; choosing *both*, the best cell. Per state: $v_0 = \text{mean}_{p,u} Y$; $v_P = \max_p \text{mean}_u Y$; $v_U = \max_u \text{mean}_p Y$; $v_{PU} = \max_{p,u} Y$. The primary contrast per task is $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$: positive means contact selection is worth more than direction selection, negative the reverse. Secondary estimands are the inter-

Table 1. Experimental-unit hierarchy: state → table → action cell, replicated over candidates. Any cell failure or re-placement violation invalidates the whole table (quarantined, never averaged in); analysis uses only valid C0 tables.

Unit	Count
Canonicalized states (16 per task)	48
Candidate execution regimes (C0–C3)	4
State–candidate tables (48 × 4)	192
Action cells per table (8 × 8)	64
Executed action cells	12,288
Invalid tables (C0/C1/C2/C3: 8/10/16/8)	42
Valid C0 tables analyzed (12/13/15 per task)	40
Analyzed action cells (40 ×64)	2,560

action $I = \mathbb{E}_s[v_{PU} - v_P - v_U + v_0]$ (joint value neither marginal explains), oracle gain $\mathbb{E}_s[v_{PU} - v_0]$, and two-way ANOVA variance shares. As maxima, v_P, v_U, v_{PU} are upward-biased under noise and I inherits a positive bias; a zero-displacement control (identical primitive, zero motion; $|Y|$ median $1.4{\times}10^{-3}$, p99 $3.1{\times}10^{-3}$) sets the floor against which we read I.

3.4. Validity, statistics, and deviations

A table is valid only if all 64 cells complete with successful grasp, converged settle, finite outputs, and re-placement error of the canonicalized state within the 10^{-4} m reset guard; any failure invalidates the whole table (one retry, no cell deletion, no post-hoc state replacement). Invalid tables are reported separately from valid null results. Guards cover no-op drift, repeatability (top-action equivalence, tie tolerance 10^{-4}), settle convergence, non-finite incidents, invalid-table rate, and factor-dependent invalidity gaps.

Uncertainty uses a stratified (task × initial-shape) cluster bootstrap over states with 10^4 resamples (Efron & Tibshirani, 1993); the three per-task primary contrasts form one family with studentized max-t simultaneous 95% intervals. The preregistered minimum absolute effect is 0.02; a task is labeled *contact-dominant* only if $\Delta_{WH} \geq 0.02$ with simultaneous lower bound > 0, *direction-dominant* symmetrically, else *balanced/inconclusive*. Cross-task contrasts use the shared strata {u_bend, s_curve, random_smooth} with equal weights (a composition-confound correction) and are exploratory.

Deviations from preregistration. (1) 12/13/15 valid development-split states per task instead of the registered 96/64; (2) hidden-split confirmation and clean reproduction were not executed; (3) cross-task contrasts use the shared-strata correction above; (4) claim language is narrowed to contact-location versus fixed-magnitude planar-direction choice; (5) the registered *study-level* placement guard is operationalized per table: states that cannot be re-placed within 10^{-4} m yield invalid, quarantined tables rather than aborting the study. Grid, candidates, priority order, eligibility, guards, and estimands are unchanged.

Table 2. Primary contrast $\Delta_{WH} = \mathbb{E}_s[v_P - v_U]$ per task (candidate C0); studentized max-t simultaneous 95% CIs over the three-task family; n = valid states. Labels apply the preregistered rule (minimum effect 0.02) *descriptively*; development split only.

Task	n	Δ_{WH}	simult. 95% CI	label
t1a straighten	12	+0.041	[+0.029, +0.053]	contact-dominant
t1b single bend	13	-0.034	[-0.073, +0.005]	balanced/inconclusive
t1c endpoint reposition	15	-0.130	[-0.161, -0.100]	direction-dominant

Table 3. Factorial decomposition per task (candidate C0; state means, pointwise 95% bootstrap CIs). I is upward-biased by noise; compare with the zero-displacement floor (median $1.4{\times}10^{-3}$, p99 $3.1{\times}10^{-3}$).

Task	v_0	$v_P{-}v_0$	$v_U{-}v_0$	I	$v_{PU}{-}v_0$
t1a straighten	-0.151	+0.078	+0.037	+0.228	+0.343
t1b single bend	-0.142	+0.104	+0.139	+0.154	+0.397
t1c endpoint reposition	-0.066	+0.083	+0.214	+0.007	+0.304

4. Results

Qualification. No candidate execution regime passed the preregistered invalid-table-rate guard; every number below is descriptive development-split pilot evidence. The invalid-table rate guard fails: 42 of 192 tables are invalid (rate $0.219 > 0.02$), dominated by 6 states with a canonicalization re-placement artifact (error up to $6.4{\times}10^{-3}$); the registered 0.02 threshold admits less than one invalid table per candidate at this pilot scale, so under the strict fixed-priority rule no candidate regime is selectable. Invalid tables are quarantined, never averaged in, and listed in the released data. The full-state-set no-op p99 guard also fails, driven entirely by quarantined states; restricted to analyzed states, no-op drift is median $1.0{\times}10^{-8}$, p99 $1.5{\times}10^{-4}$, well within thresholds. The failure sits in the execution substrate, not the hypothesis; quarantine kept the artifact out of the estimates. Signal-level controls on the analyzed tables pass: repeat top-action equivalence 1.000 (≥ 0.99), settle convergence 0.9957 (≥ 0.95), non-finite incidents 0, factor-dependent invalidity gap $9.8{\times}10^{-3}$, placement error over analyzed tables $7.8{\times}10^{-5}$ ($\leq 10^{-4}$). Throughput was 5.42 valid action cells/s at 512 parallel environments.

Primary contrast: factor importance reverses across tasks. Table 2 and Figure 2 give Δ_{WH} per task. Applying the preregistered labeling rule descriptively, t1a straighten is labeled *contact-dominant*; t1b single bend is labeled *balanced/inconclusive*; t1c endpoint reposition is labeled *direction-dominant* on this pilot. Figure 1 shows why: straightening (t1a) tables vary far more across contact rows than direction columns—once a good grasp point is chosen, many pull directions work; endpoint repositioning (t1c) shows the reverse column striping. Across all 4 execution

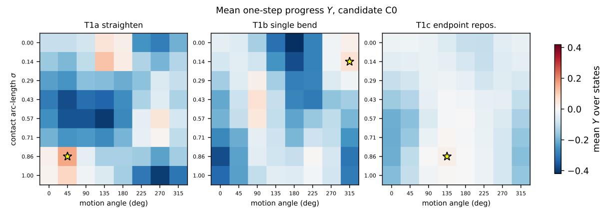

Figure 1. Mean one-step progress Y per task (candidate C0). Rows: contact arc-length σ; columns: motion angle. Star: best mean cell.

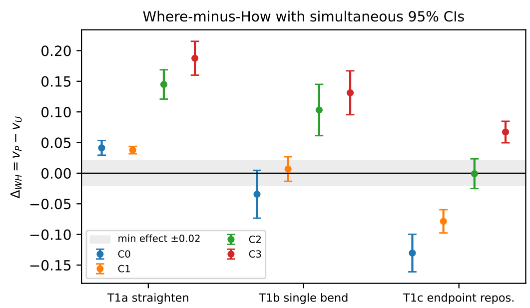

Figure 2. Δ_{WH} per task and candidate with simultaneous 95% intervals; shaded band is the preregistered minimum effect ± 0.02.

regimes (C0–C3), the sign of Δ_{WH} per task is *not* consistent (Figure 2); non-primary candidates are descriptive only.

Decomposition and interaction. Table 3 reports marginal and joint components. Interaction estimates are t1a straighten: +0.228 [+0.217, +0.239]; t1b single bend: +0.154 [+0.138, +0.168]; t1c endpoint reposition: +0.007 [-0.007, +0.020]. The t1a and t1b estimates sit far above the zero-displacement noise floor; the t1c interval includes zero. Variance shares of the mean tables attribute contact/motion/interaction variance as t1a straighten: 12%/2%/86%; t1b single bend: 8%/16%/76%; t1c endpoint reposition: 12%/66%/21%. For t1a/t1b the interaction rivals or exceeds either main effect—the best cell is a specific contact–direction *pairing* neither marginal predicts; for t1c it is negligible. On the shared strata, the exploratory t1c−t1a interaction difference (preregistered as H3, recomputed under the shared-strata correction) is -0.180 [-0.201, -0.160]. The interval excludes zero.

5. Discussion and limitations

Interpreting direction dominance. A negative Δ_{WH} need not mean direction *knowledge* is constructively more valuable: a wrong direction actively harms ($Y < 0$) while a wrong contact is often merely unhelpful ($Y \approx 0$), so direction dominance can reflect harm avoidance; the sign of v_0 (grand mean t1a straighten: -0.151; t1b single bend: -0.142; t1c endpoint reposition: -0.066) bounds this share, and the expressiveness asymmetry of Section 1 biases the same way.

Lessons for action parameterization design. First, *equal cardinality does not equalize factors*: the contact factor spans the rope's full arc length, the motion factor a thin slice of motion space; attribution studies should publish such an expressiveness audit (registered follow-ups: a 16×16 probe, a magnitude-augmented motion factor). Second, *factor importance is a property of the task*: the sign of Δ_{WH} reverses between straightening and endpoint repositioning. Architectures that hard-code priority for one factor—dense where-maps with a coarse motion head (Mo et al., 2021; Wu et al., 2023; Zhou et al., 2023)—inherit a task-dependent bias; sizable interactions on two of three tasks argue for retaining joint contact–motion scoring.

Lessons for simulator benchmark design. (i) *Reset fidelity is a first-class property*: without per-table placement checks the artifact would have silently biased every estimate; simulators should expose and test deterministic replacement (Lin et al., 2021; Cao et al., 2026). (ii) *Max-statistics need noise-floor controls*: best-of-table quantities are upward-biased, so reported oracle or best-action gaps need a sham control. (iii) *Guard thresholds must anticipate pilot scale*: a 0.02 invalid-rate threshold admits fewer than one invalid table per candidate here, so one localized artifact makes the guard unsatisfiable; thresholds should scale with sample size or carry explicit quarantine semantics. We downgraded all claims rather than relax the threshold post hoc; effect-blind selection makes the downgrade credible rather than convenient (Nosek et al., 2018; Agarwal et al., 2021).

Limitations. This is a pilot: 12/13/15 states per task on the development split, one simulator, one rope parameterization, no hidden-split confirmation, no clean reproduction, no grasp realism. Effects are one-step greedy quantities on canonicalized quasi-static states and do not license multi-step policy conclusions; the motion factor omits magnitude, vertical, and rotational components. Candidate-level evidence is unstable: the sign of Δ_{WH} is not consistent across regimes (Figure 2), so labels are regime-specific. Wide intervals mean *balanced/inconclusive* reads as underpowered, not as evidence of equality.

6. Conclusion

We contribute a preregistered, guard-audited pilot diagnostic attributing one-step DLO progress to contact-location versus fixed-magnitude planar-direction choice. Descriptively, contact choice mattered most for straightening, direction choice for endpoint repositioning, with substantial interactions on two of three tasks; because no candidate regime passed the preregistered validity guard, these are pilot observations, not confirmed findings. The design scales beyond the pilot; per-cell raw data, configs, and seeds are released.

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